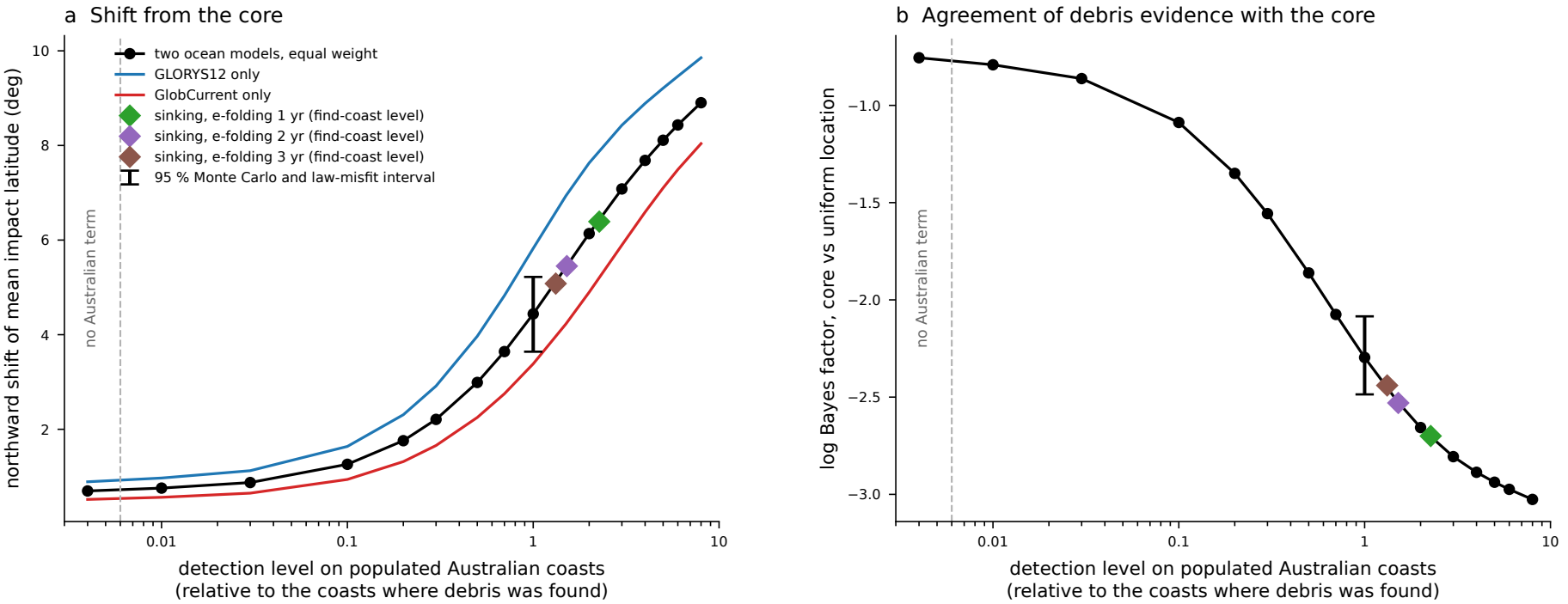
Effect of no debris found on populated Australian coasts, by assumed detection level



Plain English. No aircraft debris was found on the populated coasts of Australia. The more likely it is that people would find and report debris there, the more this pushes the impact point north, away from the impact point given by the flight-path calculation (the core). The point left of the dashed line has no Australian effect. Detection level 1 means populated Australian coasts are as likely to report debris as the coasts where debris was found. The error bar shows the noise from the drift calculation and from the fitted formula. Diamonds: if floating debris sinks over time (for example from marine growth), debris that reaches Australia early is affected less than debris that reaches Africa later; this has the same effect as a higher detection level. A negative log Bayes factor means the debris evidence agrees less with the core than with an impact at a uniform location along the arc.

Technical. Run J waypoints, 00:19 option D0' ('D0 total q=0.333 (C2 silent)', 24 seeds); drift run A surfaces rebuilt (GLORYS12V1 0.494 m + ERA5; GlobCurrent 0 m + ERA5, windage U(-0.75,-0.60) % U10), h = 50 km, L' = L exp(Delta), Delta = -k ln(1 + rho W_pop / (c F)) per model (k, c fitted to pilot options B and C on the 178-node 15-160 E pilot; GLORYS12 k 5.47 c 0.388; GlobCurrent k 4.60 c 0.366), W_pop = city + inner-regional segments (ABS RA 2021), IDW (8 nearest) to all nodes. Populated-coast term is post hoc and reconstructed: labelled sensitivity. Error bar at rho = 1: 200 replicates, node noise N(-s^2/2, s) by 1-deg band + pilot Delta noise (split-half SD 0.50/0.49) + law misfit (SD 0.62/0.49). Sinking: survival exp(-t/tau) on first-beaching day to 30 Jun 2016 (snapshot harness, core PDF draws); rho_eff = ratio of mean survival, all Australian coasts vs the five find regions; k, c held fixed (approximation).